Project Name: Manuzam
Developer: Muhammad Zain ul Abideen

Sprint 2 Summary – Backend & Frontend Development

In Sprint 2, the primary focus was on developing the system's core functionalities, including the user interface, backend services, and the integration between them. The following key tasks were completed:

Completed Issues

- **Issue #1: Create the Frontend**
  - Built a clean, responsive frontend with a user-friendly interface for uploading audio files.
  - Used Axios to send data to the backend, handle responses, and User authentication.

- **Issue #2: The Backend APIs**
  - Developed RESTful endpoints using **FastAPI** to handle user and machine profiles, and audio uploads.
  - Added CORS middleware to allow communication between the frontend and backend.

- **Issue #4: Connect Frontend with Backend ML Result**
  - Established the communication flow from the frontend to the backend.
  - Handled API integration and response parsing for future ML results.

All completed issues are tracked on the project GitLab repository.

Ongoing Work (Backlog for Sprint 3)

- **Issue #3: Implement Initial ML Pipeline**
  - This task is in progress and will focus on training a basic fault classification model.

Milestone 2 Summary and Transition to Milestone 3

Milestone 2 was completed and officially closed on June 11th. This sprint focused on establishing the core functionality of the system.

Following the closure of Milestone 2, **Milestone 3** has officially begun on the same day (June 11th). This sprint, spanning the next two weeks, will concentrate on:

- Training the machine learning model and implementing audio fingerprinting techniques.
- Improving the audio file upload system.
- Finalising the integration of all system components to ensure a unified workflow.

Milestone 4 will follow as the final sprint, aimed at delivering a fully functional and tested application.